

# A Finite Cutoff-Sensitivity Obstruction for the V59 Residual Phase

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## Abstract

TPC-267 found a finite quarter contraction for a literal V59 residual using the coarse local comparison cutoff  $z = 2$ . We perform the next hostile test while keeping the prime shell, unit masks, deleted diagonal, source coefficient, kernel operator, and rank-three projection fixed. Outward rational intervals classify sixteen finite rows. Six matched  $z = 2$  controls contract below  $1/4$ , while six perturbed rows are certified above  $1/4$ . The same central clock  $(N, H, Q, s) = (64, 15, 4, 1)$  changes from  $\rho \leq 0.2320126753$  at  $z = 2$  to  $\rho \geq 0.2735126949$  at  $z = 3$ . This refutes a universal quarter claim over the declared finite parameter family, but it is not an asymptotic counterexample and pays no fixed-power credit.

## 1 Position and claim firewall

The typed end-to-end audit of TPC-266 requires an actual growing V59 radius or signed-phase estimate before the endpoint budget can be paid. TPC-267 supplied a reproducible finite interface and twelve  $z = 2$  rows with  $|C_\perp|/R < 1/4$ , while explicitly leaving the growing radius and source-specified profile open. The present paper asks whether that finite inequality is stable under the interface choices declared as modeling choices.

CUTOFF\_SENSITIVITY\_OBSTRUCTION  
*numerically certified finite result.*

## 2 The finite physical operator

Let  $N$  be even and set

$$\mathcal{I}_N = \{N/2 + 1, \dots, N\}, \quad \mathcal{Q}_Q = \{q : q \text{ prime}, Q < q \leq 2Q\}.$$

For an integer  $z \geq 2$ , define

$$b_N^{(z)}(u) = C_2^{(z)} 1_{2 \nmid u} \prod_{p \leq z} \frac{p}{p-1} \prod_{\substack{p|u \\ p > z}} \frac{p-1}{p-2}, \quad (1)$$

with value zero when  $u+2$  has a prime factor at most  $z$ , and

$$C_2^{(z)} = \prod_{p > z} \left(1 - \frac{1}{(p-1)^2}\right).$$

The source coefficient and residual input are

$$\beta_N(t) = \frac{\Lambda(t)}{\log t} - \sum_{\substack{d|t \\ d^{400} \leq N^{133}}} \mu(d), \quad w_N(u) = \Lambda(u+2) - b_N^{(z)}(u). \quad (2)$$

For  $s \in \{1, 2, 3\}$ , use

$$K_{H,s}(h) = \left(1 + (h/H)^2\right)^{-s}, \quad (3)$$

and the literal centered operator

$$A(u, t) = 1_{u \neq t} \sum_{q \in \mathcal{Q}_Q} q K_{H,s}(u-t) 1_{q|ut} \left(1_{u \equiv t \pmod{q}} - \frac{1}{q-1}\right). \quad (4)$$

The prime-only shell, both unit masks, and the deleted diagonal are therefore part of every row. Put  $g = A\beta_N$ .

### 3 Projection and interval decision

Partition  $\mathcal{I}_N$  into four consecutive blocks of size  $B = N/8$ . Use the three orthogonal contrasts

$$c_0 = (1, 1, -1, -1), \quad c_1 = (1, -1, 0, 0), \quad c_2 = (0, 0, 1, -1). \quad (5)$$

For block sums  $W_j, G_j$ , define

$$C = \langle w_N, g \rangle, \quad C_3 = \frac{W_0 G_0}{4B} + \frac{W_1 G_1}{2B} + \frac{W_2 G_2}{2B}, \quad C_\perp = C - C_3, \quad (6)$$

and

$$R^2 = \|(I - P_3)w_N\|^2 \|(I - P_3)g\|^2, \quad \rho^2 = \frac{|C_\perp|^2}{R^2}. \quad (7)$$

**Proposition 1** (finite identity). *For every listed row, (6) is the exact cross-Gram decomposition associated with the rank-three block-contrast projection, and  $R^2 > 0$ .*

*Proof.* The three vectors in (5) have squared norms  $4B, 2B, 2B$  and are pairwise orthogonal. The orthogonal projection formula gives (6), and subtracting the projected cross-Gram gives  $C_\perp$ . The two residual norm factors are obtained by applying the same formula to each squared norm. Positivity is enclosed by the interval computation described below.  $\square$

The finite Euler product is multiplied through  $P = 50000$ . The tail uses

$$\prod_{p > P} \left(1 - \frac{1}{(p-1)^2}\right) \geq 1 - \sum_{p > P} \frac{1}{(p-1)^2} > 1 - \frac{1}{P-1}. \quad (8)$$

The logarithms are evaluated at 100 decimal digits with a  $10^{-25}$  guard. Every endpoint is then rounded outward to a rational grid of width  $10^{-30}$ . Thus the decision in (7) uses only rational interval operations:

$$\rho_{\text{hi}}^2 < \frac{1}{16} \Rightarrow \text{contraction}, \quad \rho_{\text{lo}}^2 > \frac{1}{16} \Rightarrow \text{obstruction}. \quad (9)$$

Table 1: Selected rows. Displayed  $\rho$  values are rounded square-root upper endpoints from the certificate; the contraction/obstruction decision uses the separately stored interval for  $\rho^2$ .

| N   | H  | Q | s | z | displayed $\rho$ | class    |
|-----|----|---|---|---|------------------|----------|
| 64  | 15 | 4 | 1 | 2 | 0.2320126753     | contract |
| 96  | 20 | 5 | 1 | 2 | 0.2138888503     | contract |
| 128 | 24 | 5 | 1 | 2 | 0.1267919366     | contract |
| 192 | 32 | 6 | 1 | 2 | 0.0066519390     | contract |
| 256 | 38 | 6 | 1 | 2 | 0.0093009287     | contract |
| 384 | 50 | 7 | 1 | 2 | 0.0631257787     | contract |
| 64  | 13 | 4 | 1 | 3 | 0.2689314328     | obstruct |
| 64  | 15 | 4 | 1 | 3 | 0.2735358969     | obstruct |
| 64  | 17 | 4 | 1 | 3 | 0.2758624715     | obstruct |
| 96  | 20 | 5 | 1 | 3 | 0.2996937679     | obstruct |
| 96  | 20 | 5 | 2 | 3 | 0.2886875882     | obstruct |
| 64  | 15 | 4 | 1 | 5 | 0.3851247936     | obstruct |

## 4 The finite obstruction

**Theorem 1** (cutoff sensitivity). *Among the sixteen rows in the certificate, ten are certified contractions and six are certified obstructions. In particular,*

| $(N, H, Q, s, z)$   | $\rho^2$ interval                    | classification |
|---------------------|--------------------------------------|----------------|
| $(64, 15, 4, 1, 2)$ | $[0.0538214595269, 0.0538298814898]$ | contraction    |
| $(64, 15, 4, 1, 3)$ | $[0.0748091943191, 0.0748218869170]$ | obstruction    |

so changing only the finite cutoff from  $z = 2$  to  $z = 3$  flips the quarter classification. The  $z = 3$  obstruction persists at  $H = 13$  and  $H = 17$ , and the row  $(64, 15, 4, 1, 5)$  has  $\rho \leq 0.3851247936$ .

*Proof.* The producer enumerates (1)–(4), computes the exact projection split, and propagates the outward intervals through (7). The lower endpoints of the six obstruction rows are all larger than  $1/16$ , while the upper endpoints of the ten contraction rows are all smaller than  $1/16$ . The independent replay recomputes the same sixteen ratios without importing the producer. The central pair and the clock-neighborhood rows are therefore separated from the threshold by certified intervals, not by rounded display values.  $\square$

## 5 Interpretation

The central pair gives a clean finite sensitivity statement. The physical matrix, source coefficient, clock, shell, masks, and projection are identical in the two rows; only the declared local comparison cutoff changes. The interval for the first row lies below  $1/16$ , whereas the lower endpoint for the second row is

$$0.0748091943191 > 1/16 = 0.0625.$$

Consequently a universal quarter-sector lemma over this tested finite parameter family is false. The clock perturbations  $H = 13$  and  $H = 17$  show that the  $z = 3$  obstruction is not a single rounded-clock accident, and the  $z = 5$  row supplies a stronger finite stress point. These are finite statements about the explicitly enumerated rows.

There are two important qualifications. First,  $z$  is a modeling parameter here. The source-level V59 construction has a growing local cutoff, and this paper does not prove that its finite values are represented by any fixed  $z$  in the table. Thus the result is not a source-level growing- $x$  counterexample. Second, the kernel exponent is not monotone in this diagnostic: the  $N = 64, z = 3$  row with  $s = 2$  is just below the quarter threshold, while the  $N = 96, z = 3$  row with  $s = 2$  is above it. A finite contraction therefore cannot be promoted by an informal smoothness or monotonicity argument.

The audit also does not pay any endpoint budget. It proves no bound for the size of  $R$ , gives no fixed-power saving, and supplies no arithmetic  $L^2$  estimate or full Gate-B closure. In the notation of the upstream firewall, the fixed-power credit remains zero and the twin-prime conclusion remains open.

## 6 Conclusion

TPC-268 turns the finite residual census into an adversarial diagnostic. A matched  $z = 2$  control and  $z = 3$  perturbation use the same finite physical operator but land on opposite sides of the quarter threshold, with six outward-certified obstructions across the sixteen-row family. The natural next question is not whether the most favorable finite profile can be made to contract, but whether a growing cutoff and a source-compatible smooth profile admit a uniform interval theorem. Until that question is answered, the observed phase contraction is evidence about a finite interface rather than progress toward the asymptotic twin-prime estimate.

## References

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